

Strict Scope Boundaries & Unspecified Parameters: Per the primary source document, timeline delivery dates, story point sizing, and sprint velocity numbers are marked strictly as *"Not specified in the source document"*. Future extensions (Nutrition tracking, BMI calculation, Wearable device integration, Workout reminders, Fitness analytics dashboards) are strictly excluded from baseline implementation.

1. SPRINT CADENCE & VELOCITY GOVERNANCE

SPRINT SCHEDULE & DATES

Sprint Start Date: Not specified in the source document

Sprint End Date: Not specified in the source document

Cadence Length: Not specified in the source document

VELOCITY & SIZING

Planned Velocity: Not specified in the source document

Total Story Points: Not specified in the source document

Estimation Standard: Not specified in the source document

TEAM STRUCTURE & ROLES

Execution Roles: Lead Backend Engineer, QA Engineer, Solution Architect

User Role: Authenticated User (single actor)

2. DOCUMENTED DELIVERY & DEPENDENCY EXECUTION CHAIN

The baseline implementation strictly enforces linear dependency traversal. Downstream service layers cannot be implemented or validated until preceding infrastructural gates are verified.



3. EPIC SUMMARY & STRUCTURAL ARCHITECTURE

EPIC IDENTIFIER	EPIC DESIGNATION	SCOPE OF TECHNICAL WORK	DOCUMENTED PREDECESSOR
EPIC-01	Project Architecture	Define project directory structure, separation of concerns (MVC architecture: routes, controllers, models, middleware, config), and execution pipeline.	NONE (ROOT GATE)
EPIC-02	Project Setup and Configuration	Initialize Node.js package environment, configure dependencies (Express, Mongoose, dotenv, cors, jsonwebtoken, bcryptjs, nodemon), and load environment variables.	EPIC-01
EPIC-03	Backend Development	Implement controllers, REST route handlers, JWT authentication middleware, multi-criteria workout filtering, and Google Gemini API generative integration.	EPIC-02, EPIC-04
EPIC-04	Database Configuration	Configure MongoDB connection using Mongoose, establish database connection lifecycle, and define strict schemas for Users and Workouts.	EPIC-02
EPIC-05	Project Execution & API Testing	Execute progressive API verification using Postman / ThunderClient following the mandatory sequence: Authentication → Workouts → AI Intelligence.	EPIC-03

4. PRODUCT BACKLOG MATRIX: TECHNICAL SPECIFICATIONS (EPICS 1 – 3)

Comprehensive backlog matrix tracking architectural setup, database scaffolding, and authentication services against strict technical constraints.

BACKLOG ID	EPIC	FEATURE / TASK	TECHNICAL SCOPE & IMPLEMENTATION DESCRIPTION	PRIORITY	DEPENDENCY	STATUS
PBI-01	Epic 1: Architecture	Directory & Layout Scaffolding	Establish clean MVC codebase modularization: /controllers, /models, /routes, /middleware, /config, and root server.js.	CRITICAL	None	NOT SPECIFIED
PBI-02	Epic 2: Setup & Config	Runtime & Manifest Setup	Initialize Node.js runtime environment (v16+) and configure root package.json with production and developer dependencies.	CRITICAL	PBI-01	NOT SPECIFIED
PBI-03	Epic 2: Setup & Config	Environment Configuration	Implement dotenv configuration to securely inject environment variables: PORT, MONGO_URI, JWT_SECRET, and GEMINI_API_KEY.	CRITICAL	PBI-02	NOT SPECIFIED
PBI-04	Epic 4: Database Config	Mongoose Connection Scaffolding	Create database connection module in /config/db.js using Mongoose to asynchronously bind to MongoDB via MONGO_URI.	CRITICAL	PBI-03	NOT SPECIFIED
PBI-05	Epic 4: Database Config	User Schema Modeling	Define Mongoose User schema with strict field validations: name (required string), email (required unique string), and password (required hashed string).	CRITICAL	PBI-04	NOT SPECIFIED
PBI-06	Epic 4: Database Config	Workout Schema Modeling	Define Mongoose Workout schema: userId (ObjectId ref User), name (required), category, duration (number), caloriesBurned, and date.	CRITICAL	PBI-04 , PBI-05	NOT SPECIFIED
PBI-07	Epic 3: Backend Dev	User Registration Engine	Implement POST /api/auth/register with bcrypt.js password hashing, duplicate email validation, and JWT token issuance.	CRITICAL	PBI-05	NOT SPECIFIED
PBI-08	Epic 3: Backend Dev	User Login & Token Issuance	Implement POST /api/auth/login verifying bcrypt password hashes and signing JSON Web Tokens with JWT_SECRET.	CRITICAL	PBI-05 , PBI-07	NOT SPECIFIED
PBI-09	Epic 3: Backend Dev	Auth Guard Middleware	Develop Bearer JWT verification middleware to extract token from headers, verify cryptographic signature, and inject authenticated req.user.	CRITICAL	PBI-08	NOT SPECIFIED
PBI-10	Epic 3: Backend Dev	User Profile Endpoint	Implement GET /api/auth/profile secured by Auth Guard to return authenticated user record excluding hashed password.	HIGH	PBI-09	NOT SPECIFIED

5. PRODUCT BACKLOG MATRIX: TECHNICAL SPECIFICATIONS (EPICS 3 & 5)

Continuation of product backlog items covering workout lifecycle operations, search filters, AI integration, and API testing execution.

BACKLOG ID	EPIC	FEATURE / TASK	TECHNICAL SCOPE & IMPLEMENTATION DESCRIPTION	PRIORITY	DEPENDENCY	STATUS
PBI-11	Epic 3: Backend Dev	Workout Creation Endpoint	Implement POST /api/workouts to persist workout records bound to authenticated user's ID via JWT context.	CRITICAL	PBI-06 , PBI-09	NOT SPECIFIED
PBI-12	Epic 3: Backend Dev	Workout Retrieval Endpoints	Implement GET /api/workouts and GET /api/workouts/:id ensuring users only access their own workout documents.	CRITICAL	PBI-11	NOT SPECIFIED
PBI-13	Epic 3: Backend Dev	Workout Update & Deletion	Implement PUT /api/workouts/:id and DELETE /api/workouts/:id enforcing tenant isolation and record ownership checks.	HIGH	PBI-12	NOT SPECIFIED
PBI-14	Epic 3: Backend Dev	Search & Filter Query Engine	Implement multi-criteria filtering on GET /api/workouts supporting query params: name, category, and date.	HIGH	PBI-12	NOT SPECIFIED
PBI-15	Epic 3: Backend Dev	AI Regimen Recommendation	Implement POST /api/ai/recommend connecting to Google Gemini API to return personalized workout plans from user input parameters.	HIGH	PBI-09	NOT SPECIFIED
PBI-16	Epic 3: Backend Dev	AI Historical Insights Engine	Implement GET /api/ai/insights to extract user workout history from MongoDB, construct AI prompt, and return Gemini analytics.	HIGH	PBI-12 , PBI-15	NOT SPECIFIED
PBI-17	Epic 5: Execution & Test	Auth & Profile API Testing	Execute Step 1 of testing: verify register, duplicate check, login, token issuance, and profile retrieval via Postman / ThunderClient.	CRITICAL	PBI-08 , PBI-10	NOT SPECIFIED
PBI-18	Epic 5: Execution & Test	Workout Management API Testing	Execute Step 2 of testing: verify workout creation, retrieval, updates, deletions, and multi-criteria query search execution.	CRITICAL	PBI-13 , PBI-14 , PBI-17	NOT SPECIFIED
PBI-19	Epic 5: Execution & Test	AI Intelligence API Testing	Execute Step 3 of testing: verify Gemini recommendation endpoint, historical workout insights aggregation, and error fallback resilience.	HIGH	PBI-15 , PBI-16 , PBI-18	NOT SPECIFIED

6. SPRINT GOVERNANCE & PLANNING AUDIT

PLANNING METRIC	DOCUMENTED VALUE	COMPLIANCE & SOURCING NOTE
Total Epics Documented	5 Epics	Architecture, Setup & Config, Backend Dev, Database Config, Execution & Testing.
Total Planned Backlog Items	19 Product Backlog Items	Spanning architectural foundation through full verification suite.
Sprint Story Point Sum	NOT SPECIFIED IN SOURCE DOC	No story point estimation system or point values exist in the source document.